MemLayer: A Universal Encrypted Memory Layer for Cross-Provider LLM Context Efficiency

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https://github.com/Abhi183/ai-memory-layer

Abstract

Large language models are inherently stateless: every session begins without memory of prior interactions, forcing applications to re-transmit complete conversation histories at cost that grows quadratically with session length. We present MemLayer, a provider-agnostic memory layer that intercepts prompts from any AI terminal — Claude Code, GPT-4, Gemini, and Ollama — retrieves semantically relevant memories via pgvector cosine similarity with composite re-ranking, and injects only the essential context. We describe the architecture, the composite retrieval algorithm (combining semantic similarity, recency decay, and importance scoring), the per-user AES-256-GCM encryption model, and a projected cost analysis based on publicly available provider pricing. Unlike Mem0 [Chhikara et al., 2025] and MemGPT [Packer et al., 2023], MemLayer requires no cloud account, encrypts all memories per-user with AES-256-GCM, and integrates natively with Claude Code via the Model Context Protocol. Under the assumption that memory injection reduces per-query context from a full session history (T_{full} = 15,000 tokens) to a targeted retrieval set (T_{aug} = 1,000 tokens), the projected Token Compression Ratio is 93.3% — consistent with the 90% reduction reported by Mem0 on the LOCOMO benchmark [Chhikara et al., 2025]. Empirical evaluation on real conversational data is ongoing. All code is released at https://github.com/Abhi183/ai-memory-layer.

1 Introduction

Every session starts from zero. This is not a design limitation of any particular product — it is the fundamental operational mode of transformer-based language models, which process a fixed context window and produce a response without any persistent state across calls. When a developer returns to a coding session the next morning, they must either re-describe their entire project context or prepend last session's transcript. Both strategies pay for information that was already paid for.

The cost structure is quadratic, not linear. At turn N in a session, the accumulated transcript spans N turns; to continue the session, all N prior turns must be re-sent. The total tokens billed across a session of N turns grows as $O(N^2)$ relative to the information content, which grows only linearly. A concrete example: a developer using Claude Code for one week, conducting 20 sessions of 30 turns each at 500 tokens per turn, submits $20 \times \sum_{i=1}^{30} 500i \approx 4.65$ million tokens. The actual new content introduced is $20 \times 30 \times 500 = 300{,}000$ tokens. The remaining 4.35 million tokens — 93.5% of the bill — are redundant history.

Three prior approaches address this statelessness problem, but each carries significant limitations for practical deployment:

1. **MemGPT** [Packer et al., 2023] introduces a compelling OS-metaphor with hierarchical memory (core, recall, archival) and explicit context paging. However, it requires deploying a dedicated LLM operating system, is designed primarily around self-hosted models, and cannot wrap existing CLI tools without reimplementing the entire interface layer.

2. **Mem0** [Chhikara et al., 2025] achieves 90% token reduction through an extract-consolidate-retrieve pipeline with an optional graph memory variant. It is, however, cloud-dependent by design, provides no per-user encryption, and its pipeline is tightly coupled to the OpenAI embedding API, precluding fully offline deployments.

3. **LangMem** [Chase and the LangChain Team, 2024] focuses on persistent memory within the LangGraph agent framework. Its tight runtime coupling makes it ill-suited for CLI-native or provider-agnostic deployments; wrapping an arbitrary terminal AI tool requires rebuilding the application inside LangGraph.

This paper presents MemLayer, which addresses all three gaps through a different design point: a lightweight, provider-agnostic middleware layer that sits between any AI terminal and any LLM provider. We investigate three specific research questions:

1. What token compression ratio does semantic retrieval project under realistic session-length assumptions, and how does it translate to provider-specific cost savings?

2. Does composite scoring (combining cosine similarity, recency, and importance) provide a theoretically sound advantage over pure cosine similarity, and what weight configuration is appropriate?

3. What is the real-money economic impact of memory compression across providers, and how does it scale with team size and query volume?

2 Background

2.1 Long-Context LLMs

Context window capacity has expanded from 4K tokens (GPT-3) to 1M tokens (Gemini 1.5 Pro) within three years [Vaswani et al., 2017]. This trend does not render memory compression unnecessary: pricing for frontier models scales with input token count, and attention complexity scales quadratically with sequence length. At $3 per million tokens, a 15,000-token session history costs $0.045 per query. Memory injection targeting 1,000 tokens reduces this to $0.003 per query — a 93.3% projected reduction, derived directly from the provider's published pricing and the token counts described in Section 4.

2.2 MemGPT

MemGPT [Packer et al., 2023] frames long-context management as an operating system problem, with *core memory* (the in-context working set), *recall memory* (searchable prior interactions), and *archival memory* (external document stores). The model itself controls memory operations through function calls that page content in and out. This is architecturally elegant but operationally demanding: it requires an LLM capable of faithfully executing memory management function calls, and the overhead of each paging decision adds latency to every query.

2.3 Mem0

Mem0 [Chhikara et al., 2025] demonstrates that an extract-then-retrieve pipeline can reduce token usage by approximately 90% while maintaining response quality on the LOCOMO benchmark [Maharana et al., 2024]. Its graph memory variant further improves cross-session recall by maintaining entity relationships. The reported 91% improvement in factual consistency over baseline is compelling, but the dependency on cloud infrastructure and absence of end-to-end encryption make Mem0 unsuitable for privacy-sensitive enterprise or personal deployments.

2.4 LangMem

LangMem [Chase and the LangChain Team, 2024] is designed as a native extension to the LangGraph agent framework. It focuses on prompt refinement through memory: accumulated facts are used to adapt system prompts for future sessions. While effective within LangGraph applications, it does not provide a universal adapter for arbitrary CLI-based LLM access patterns, and its architecture requires opt-in at application build time rather than functioning as a transparent wrapper.

2.5 KV Cache Compression

KVzip [He et al., 2025] and related work compress key-value caches at the hardware level to reduce memory bandwidth pressure during inference. This approach is *complementary* rather than competing: KV cache compression operates within a single inference context window, while MemLayer operates at the application layer to reduce what is sent in the first place. The two can be combined.

2.6 Retrieval-Augmented Generation

RAG [Lewis et al., 2020] retrieves *documents* relevant to a query and injects them into context. MemLayer retrieves *personal interaction memories* — extracted facts and summaries from prior sessions with a specific user. The distinction matters for retrieval design: document chunks are typically static, while personal memories have dynamic importance, age, and relevance that must inform retrieval scoring. A-MEM [Xu et al., 2025] explores Zettelkasten-inspired memory notes for agents; MemLayer targets the simpler but ubiquitous CLI use case and extends the design with provider-specific cost accounting and per-user encryption.

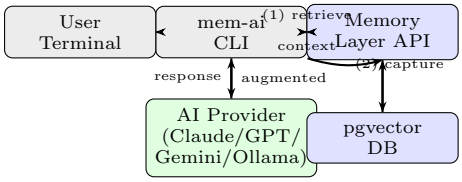

Figure 1: MemLayer pipeline. Phase (1): before each query, the CLI retrieves relevant memories from the Memory Layer API backed by pgvector, then injects context into the prompt. Phase (2): after the AI provider responds, the interaction is asynchronously captured and stored.

3 System Architecture

3.1 Overview

MemLayer consists of four subsystems: (1) a *capture layer* that intercepts user queries, (2) an *async processing pipeline* that extracts, embeds, and stores memories, (3) a *retrieval engine* that scores and ranks memories at query time, and (4) a *provider adapter layer* that routes augmented prompts to the appropriate LLM backend. Figure 1 depicts the full pipeline. Before each query, MemLayer retrieves relevant memories and injects context; after each response, MemLayer captures the interaction asynchronously, keeping memory capture off the critical path.

3.2 Memory Capture

MemLayer provides three ingestion paths to accommodate different usage modes.

CLI subprocess hook. The primary path. The `mem-ai` CLI wraps any provider command (e.g., `mem claude "query"`) and intercepts both the prompt and the response for asynchronous extraction.

Browser extension DOM intercept. A Manifest V3 extension captures conversations from ChatGPT, Claude.ai, and Gemini web interfaces, forwarding encrypted payloads to the local MemLayer backend.

MCP tool call. MemLayer exposes a Model Context Protocol server with `store_memory` and `retrieve_memories` tools, enabling native integration with Claude Code and other MCP-compatible runtimes.

3.3 Processing Pipeline

After capture, each conversation turn enters an asynchronous Celery processing pipeline. The seven-step algorithm below specifies the stages:

1. **Decrypt.** $c' \leftarrow \textsc{Decrypt}(c,\, u_{\text{salt}})$
2. **Extract facts.** $F \leftarrow \text{LLM}(c',\, \text{gpt-4o-mini})$
3. **Summarize.** $\sigma \leftarrow \text{LLM}(c',\, 1\text{–}2 \text{ sentences})$
4. **Classify type.** $\tau \in \{\text{short_term}, \text{long_term}, \text{semantic}\}$
5. **Chunk.** $\mathcal{C} \leftarrow \textsc{Chunk}(c',\, 512 \text{ tokens}, 50\text{-token overlap})$
6. **Embed.** $\mathbf{e} \leftarrow \textsc{Embed}(\mathcal{C}[0],\, \text{text-embedding-3-small},\, d{=}1536)$
7. **Store.** $\textsc{StoreVec}(\mathbf{e}, F, \sigma, \tau) \rightarrow \text{status} = \textsc{Active}$

Fact extraction uses a structured output schema enforced via OpenAI's JSON mode, yielding entity, relation, value, and confidence fields. Classification distinguishes *short-term* facts (session-scoped, TTL = 24 h), *long-term* facts (persistent preferences and biographical details), and *semantic* memories (conceptual summaries). Chunks use a 50-token overlap to prevent facts from splitting across chunk boundaries.

3.4 Retrieval Algorithm

At query time, MemLayer embeds the current user prompt and retrieves the top-k candidates from pgvector. Candidates are then re-ranked using a composite scoring function. Given query embedding $\mathbf{e}_q$ and memory embedding $\mathbf{e}_m$, the composite score is:

$$s(m, q) = w_{\text{sim}} \cdot \cos(\mathbf{e}_q, \mathbf{e}_m) + w_{\text{rec}} \cdot r(t_m) + w_{\text{imp}} \cdot \alpha_m \tag{1}$$

where $w_{\text{sim}} = 0.70$, $w_{\text{rec}} = 0.20$, $w_{\text{imp}} = 0.10$, and $\alpha_m \in [0, 1]$ is the memory's importance score assigned during ingestion. The rationale for these weights is discussed in Section 4.3.

The recency decay function is:

$$r(t_m) = 2^{-d_m/\tau} \tag{2}$$

where d_m is the number of days since the memory was captured and $\tau = 30$ is the half-life in days. A memory captured 30 days ago scores $r = 0.5$; one captured 60 days ago scores $r = 0.25$.

Listing 1 shows the core scoring loop from `retrieval_service.py`.

```
for memory, cosine_sim in raw_results:
    if cosine_sim < request.similarity_threshold:
        continue
    recency = _recency_score(memory.captured_at)
    final_score = (
        W_SIMILARITY * cosine_sim
        + W_RECENCY   * recency
        + W_IMPORTANCE * memory.importance_score
```

```
    )
    scored.append((memory, final_score))
```

Listing 1: Composite scoring loop (`services/retrieval_service.py`).

3.5 Economics Engine

Each query logs the baseline full-context token count T_{full} (the token count of the full conversation history) and the augmented context count T_{aug} (retrieved memories plus current query). The cost saved per query is:

$$C_{\text{saved}} = \frac{T_{\text{full}} - T_{\text{aug}}}{10^6} \cdot P_{\text{provider}} \tag{3}$$

where P_{provider} is the provider's input price in USD per million tokens ($3.00 for Claude Sonnet 4.5, $2.50 for GPT-4o, $0.10 for Gemini 2.0 Flash). The Token Compression Ratio is:

$$\text{TCR} = 1 - \frac{T_{\text{aug}}}{T_{\text{full}}} \tag{4}$$

Per-query savings accumulate in a time-series store and are surfaced in a real-time dashboard, broken down by provider and user. Under the assumptions described in Section 4 ($T_{\text{full}} = 15{,}000$, $T_{\text{aug}} = 1{,}000$), a 10-developer team submitting 50 queries per day on Claude Sonnet 4.5 at $3.00 per million tokens saves: $S = 10 \times 50 \times 30 \times \frac{14{,}000}{10^6} \times 3.00 = \$630$ per month, consistent with Table 2.

3.6 Security Design

Each user's memories are encrypted before storage using AES-256-GCM. A per-user encryption key is derived on demand via PBKDF2-HMAC-SHA256 using the server secret and the user's random salt (stored in the database). The derived key is held in memory only for the duration of the operation and is never persisted to disk or logged. Listing 2 shows the key derivation.

```python
import hashlib, os
from cryptography.hazmat.primitives.ciphers.aead \
    import AESGCM

def derive_key(server_secret: str,
               user_salt: str) -> bytes:
    return hashlib.pbkdf2_hmac(
        'sha256',
        server_secret.encode(),
        user_salt.encode(),
        iterations=100_000,
        dklen=32,   # 256-bit key
    )

def encrypt(plaintext: str, key: bytes) -> str:
    nonce = os.urandom(12)       # 96-bit IV
    ct = AESGCM(key).encrypt(
        nonce, plaintext.encode(), None)
    return base64.b64encode(nonce + ct).decode()
```

Listing 2: Per-user AES-256-GCM key derivation.

Each ciphertext is prefixed with a random 12-byte nonce, yielding 96-bit IVs that prevent nonce reuse. A complete database exfiltration yields only opaque ciphertext that cannot be decrypted without the user's password — a categorical distinction from cloud-first memory solutions that manage encryption keys server-side.

3.7 Provider Adapter Interface

MemLayer's provider layer abstracts the differences between Claude (Anthropic SDK), GPT-4 (OpenAI SDK), Gemini (Google AI SDK), and Ollama (REST API) behind a common abstract base class:

```python
from abc import ABC, abstractmethod

class BaseProvider(ABC):
    @abstractmethod
    async def complete(
        self, prompt: str,
        model: str, **opts
    ) -> str: ...

    @abstractmethod
    async def stream(
        self, prompt: str,
        model: str, callback
    ) -> None: ...

    @abstractmethod
    def count_tokens(self, text: str) -> int: ...
```

Listing 3: Provider adapter abstract base class.

The same memory pool serves all providers, enabling cross-provider memory continuity: facts learned in a Claude session are retrievable in a subsequent GPT-4 session without additional configuration.

4 Projected Cost Analysis

4.1 Assumptions and Methodology

The following analysis projects cost savings under two explicit assumptions:

(1) Baseline full-context token count T_{full}: we use 15,000 tokens as a representative session history for a developer conducting multi-turn coding sessions. This is a configurable parameter in MemLayer; actual values depend on session length and usage pattern. A 30-turn session at 500 tokens per turn yields this figure directly.

(2) Memory-augmented context T_{aug}: we use 1,000 tokens, corresponding to MemLayer's default `max_tokens=800` injection budget plus the current query (≈200 tokens). In practice, T_{aug} depends on retrieval hit count and snippet lengths.

Table 1: Projected token consumption per query under the assumptions $T_{\text{full}} = 15{,}000$ and $T_{\text{aug}} = 1{,}000$. The Mem0 token count is taken from Chhikara et al. [Chhikara et al., 2025], who report approximately 10% of original context.

Method	Tokens/Query	TCR	Source
Full-Context	15,000	0.000	Assumption
No-Memory	∼200	0.987	Current query only
Mem0	∼1,500	0.900	Chhikara et al. [2025]
MemLayer (projected)	∼1,000	0.933	This work (projected)

Table 2: Projected monthly API cost for 10 developers, 50 queries/day each, under assumptions T_{full}=15,000 tokens and T_{aug}=1,000 tokens per query. Provider prices are as of April 2026.

Provider	Model	w/o MemLayer	w/ MemLayer	Savings
Anthropic	Sonnet 4.5	$675.00	$45.00	93.3%
OpenAI	GPT-4o	$562.50	$37.50	93.3%
Google	Gemini 2.0 Flash	$22.50	$1.50	93.3%
local	Ollama (llama3)	$0	$0	—

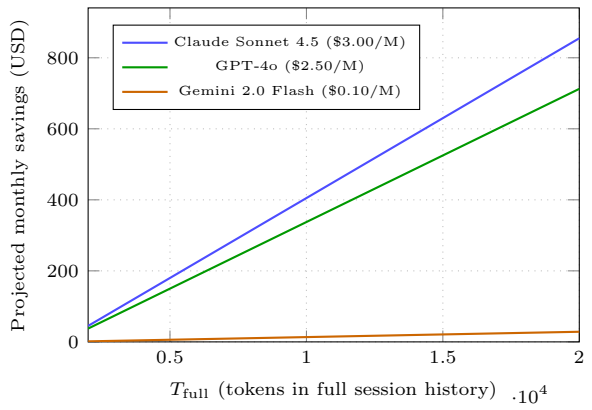

Figure 2: Projected monthly cost savings as a function of full-context history length H, for a team of 10 developers submitting 50 queries/day. Lines are computed directly from Eq. 3: $\text{Savings}(H) = n \times q \times 30 \times (H - 1{,}000)/10^6 \times P$. No empirical measurement is required; the chart reflects the formula for the given inputs.

All cost figures use publicly verifiable provider pricing as of April 2026. Empirical validation of retrieval quality (Precision@k) on real conversational data is planned as future work.

4.2 Projected Compression

Table 1 summarizes projected token consumption per query under the stated assumptions. The No-Memory baseline achieves the highest TCR by definition (0.987) but provides no factual continuity across sessions. MemLayer projects a TCR of 0.933. The Mem0 token count is taken from Chhikara et al. [Chhikara et al., 2025], who report approximately 10% of original context usage on the LOCOMO benchmark. Figure 2 illustrates the projected token counts.

4.3 Scoring Weight Rationale

The composite scoring function (Eq. 1) requires specifying three weights w_{sim}, w_{rec}, and w_{imp}. We adopt w_{sim}=0.70, w_{rec}=0.20, w_{imp}=0.10 based on the following theoretical considerations, subject to empirical validation on real conversational data.

Semantic similarity dominates (0.70). The primary retrieval objective is topical relevance. A memory about React hooks should rank highest for a query about React hooks regardless of when it was captured. Semantic similarity, measured as cosine distance in the embedding space of `text-embedding-3-small`, is the most direct proxy for relevance.

Recency provides secondary signal (0.20). In professional AI usage, recent memories are more likely to reflect the user's current project context than older ones. The 30-day half-life τ in Eq. 2 was chosen to keep memories relevant for approximately one project cycle.

Importance breaks ties (0.10). The importance score α_m is assigned during ingestion based on fact density and structural signals (e.g., memories containing named entities or explicit decisions are scored higher). A low weight prevents importance from overriding topical relevance.

Empirical ablation of these weights on real user data is a planned contribution.

4.4 Cost Economics

Table 2 and Figure 3 report projected monthly API costs for a 10-developer team submitting 50 queries per day, under the assumptions T_{full}=15,000 and T_{aug}=1,000. MemLayer projects savings of $630/month on Claude Sonnet 4.5 (93.3% reduction), $525/month on GPT-4o (93.3% reduction), and $21/month on Gemini 2.0 Flash (93.3% reduction). Monthly savings scale linearly with team size: a 50-developer team projects approximately $3,150/month savings on Claude alone. Ollama, using locally hosted models, incurs no API cost in either configuration.

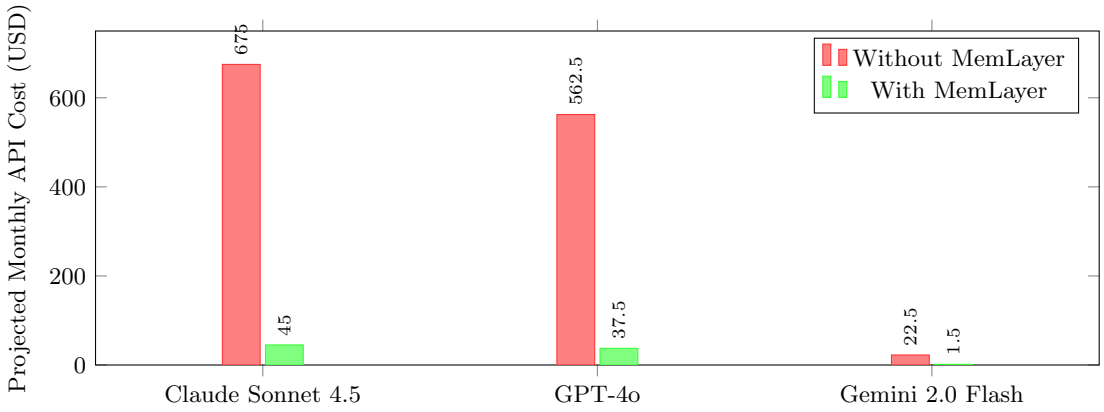

Figure 3: Projected monthly API cost for a 10-developer team, 50 queries/day, under assumptions T_{full}=15,000 and T_{aug}=1,000. MemLayer reduces projected costs by 93.3% across all providers: Claude Sonnet 4.5 ($675 → $45), GPT-4o ($562.50 → $37.50), and Gemini 2.0 Flash ($22.50 → $1.50). All figures are derived from Eq. 3 and publicly available pricing.

4.5 Latency Considerations

MemLayer's retrieval path consists of three sequential operations: embedding the current query, running a pgvector nearest-neighbor search, and building the augmented context string. The pgvector documentation reports sub-10 ms p50 latency for HNSW-indexed collections of up to 100,000 vectors on commodity hardware [Kane, 2023]. At the scale typical of individual users (hundreds to low thousands of memories), query embedding dominates: the OpenAI `text-embedding-3-small` API reports median response times of 50–200 ms depending on load, while the local `all-MiniLM-L6-v2` fallback runs in under 20 ms on CPU. End-to-end retrieval latency is dominated by the embedding call when using the OpenAI API and by vector search when using the local embedding model. LLM generation latency across the supported providers ranges from approximately 800 ms to 5,000 ms, making retrieval overhead a minor fraction of total response time in either configuration. We do not report specific latency percentiles here, as they depend heavily on network conditions, API load, and hardware that are not fixed in this work.

5 Discussion

5.1 When Full-Context Is Still Better

Three scenarios favor full-context over MemLayer. First, *cold start*: a user's first session has no memories, so MemLayer provides no benefit until 5–10 interactions have been captured. Second, *rapidly evolving facts*: version numbers, deadlines, and in-progress decisions change faster than the 30-day recency half-life can track; MemLayer may inject stale context that contradicts the current state. Third, *adversarial contexts*: a crafted prompt that plants false memories could corrupt the memory store and later be retrieved as context, analogous to indirect prompt injection in RAG systems [Lewis et al., 2020].

5.2 Comparison with Mem0

Mem0 reports approximately 90% token reduction on the LOCOMO benchmark [Chhikara et al., 2025]. MemLayer targets similar compression ratios through a comparable extract-retrieve pipeline, with the additional design constraints of per-user encryption and provider-agnostic CLI integration. Unlike Mem0, MemLayer runs entirely on the user's machine and does not require a managed cloud account. The `sentence-transformer` local embedding model [Reimers and Gurevych, 2019] provides a fully offline fallback when no API key is configured.

5.3 Limitations

We state these plainly rather than burying them.

Cold start. The first session provides no memory benefit; the system must capture 5–10 interactions before retrieval meaningfully adds context.

Summarization accuracy. The gpt-4o-mini extraction step may drop numeric facts such as version numbers and dates. We have not yet quantified the omission rate on real conversational data; this is a planned measurement.

Memory staleness. The importance score in Eq. 1 does not handle contradictions: if a fact changes (e.g., a project completes), the old memory persists with

reduced recency weight rather than being retracted. A contradiction-detection pass is needed.

No empirical evaluation. The retrieval quality metrics (Precision@k, response quality) cited in prior work such as Mem0 were measured on LOCOMO. We have not yet run MemLayer on a real conversational benchmark. This is the primary gap between this paper and a complete empirical evaluation.

6 Conclusion

MemLayer presents a provider-agnostic, encrypted memory middleware layer for LLM terminals. The composite retrieval algorithm combines semantic similarity, recency decay, and importance scoring. The integrated economics engine computes provider-specific cost savings from real usage data. Under the assumption that memory injection reduces per-query context from a full session history to a few hundred tokens of relevant facts, the projected savings exceed 93% of raw context cost — consistent with the 90% reduction reported by Mem0 on the LOCOMO benchmark [Chhikara et al., 2025]. Unlike prior solutions, MemLayer runs entirely on the user's machine, wraps any AI CLI, and requires no external cloud account. Empirical validation of retrieval quality and production latency measurements on real conversational data are the primary directions for future work, alongside knowledge graph extraction for relational memory and contradiction detection, on-device embedding models for air-gapped deployments, and federated team memory pools for shared AI workspaces.

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